HACKATHON PROJECT PRESENTATION

Serverless Notification Dashboard

Cloud-Inspired Event-Driven Architecture for Real-Time Alerts

TEAM

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PROBLEM STATEMENT

The Challenge

- Traditional notification systems require dedicated servers that are costly and hard to scale.
- Managing infrastructure for event-driven alerts adds significant operational overhead.
- Real-time updates across distributed systems are complex without a modern cloud approach.
- Developers need a simple, extensible pattern to understand serverless fundamentals.

3× More cost-efficient than server-based

∞ Scales to demand with zero config

0 Servers to manage or maintain

OUR SOLUTION

Serverless Notification Dashboard

1
User Interface
Interactive Web Dashboard

2
API Gateway
REST Event Endpoints

3
AWS Lambda
Serverless Functions

4
Amazon DynamoDB
NoSQL Datastore

Event-Driven

Notifications triggered by real-time system events

Auto-Scaling

Scales instantly with zero infrastructure changes

Cost-Efficient

Pay-per-use model, no idle server costs

Extensible

Easily add more alert types and integrations

TECHNOLOGY STACK

1

AWS Lambda

Serverless Functions

Executes backend logic triggered by events without managing servers

2

Amazon DynamoDB

NoSQL Database

Stores and retrieves notification data with millisecond latency

3

API Gateway

REST Endpoints

Exposes HTTP endpoints that trigger Lambda on incoming events

4

HTML / CSS / JS

Frontend UI

Interactive dashboard displaying real-time notification feed

5

Event-Driven Design

Architecture Pattern

Functions fire only on events — alerts, activities, system updates

6

Cloud Principles

Scalability & Cost

Demonstrates pay-per-use, auto-scaling, and zero infra management

HOW IT WORKS

System Alerts

Critical infrastructure events and warnings displayed instantly on the dashboard

User Activities

Login events, profile changes, and user interactions captured and notified in real time

Event Updates

Scheduled or triggered events pushed dynamically without page refresh

⚡ **All notifications are processed via AWS Lambda functions — no persistent server required!**

ADVANTAGES OF SERVERLESS

Cost Efficiency

Pay only for what you use. No server idle time costs. Lambda charges per millisecond of execution.

70%

Auto Scalability

Automatically handles thousands of notifications with zero configuration or manual scaling.

90%

Zero Infrastructure

No servers to provision, patch, or maintain. Focus entirely on application logic.

95%

Speed to Deploy

Faster development cycles — push code directly as functions without deployment pipelines.

80%

REAL-WORLD APPLICATIONS

This architecture can be extended to power industry-grade solutions

Monitoring Systems

Server health, uptime alerts, performance anomaly detection dashboards

Messaging Platforms

Real-time chat delivery receipts, push notifications, and in-app alerts

Alert Services

Critical incident management, on-call escalations, and emergency broadcasts

E-Commerce

Order updates, shipping notifications, inventory restock alerts

Healthcare

Patient monitoring alerts, appointment reminders, lab result notifications

Security Systems

Intrusion detection alerts, anomalous login warnings, audit trail events

THANK YOU!

Serverless Notification Dashboard

Showcasing the future of cloud-based, event-driven architecture

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